Neuromorphic Suitability of Convexity Detectors for Rural Litter Recognition

Local analysis report

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Abstract

This report analyzes three convexity-oriented image representations for a spiking neural network, also called a neuromorphic neural network, that produces an object classification after enough event patterns have been detected over a time interval. The compared methods are Radon line-contiguity, projective moment invariants, and level-set Hessian curvature. The conclusion is that Radon line-contiguity is the best mathematical fit for a spiking detector because it decomposes shape recognition into many independent line-intersection events that can be accumulated over time. Projective moments are useful as secondary global descriptors, while level-set Hessian curvature is the least natural fit because it requires dense smoothing, second derivatives, and nonlinear curvature ratios. Two concrete deployment examples are included: cigarette butt detection and soda can detection in rural environments from 10–20 m using an iPhone-style camera.

1 Detector Model

Consider a neuromorphic detector that receives image or event-camera evidence over time. The incoming event stream can be modeled as a time-indexed measure

$$\mu_t = \sum_{k:\tau_k \le t} \delta_{(x_k, y_k, \tau_k, p_k)},$$

where (x_k, y_k) is the spatial event position, τ_k is time, and p_k is polarity or channel type. For an ordinary phone camera, the same abstraction can be obtained by converting frame differences, edges, or threshold crossings into pseudo-events.

A spiking detector has pattern units indexed by j. Each unit integrates evidence through a spatiotemporal filter ψ_j:

$$z_j(t) = \int_0^t \int_\Omega \psi_j(x, y, \tau)\; d\mu(x, y, \tau).$$

The unit emits a spike when

$$z_j(t) > \theta_j.$$

For object class n, let P_n be the set of patterns relevant to that class. The object is classified after at least m required patterns have fired within a time horizon T:

$$\hat{y}(t) = n \quad \text{if} \quad \sum_{j \in P_n} \mathbf{1}[z_j(t) > \theta_j] \ge m, \qquad t \le T.$$

The key question is therefore: which geometric representation decomposes most naturally into spikeable evidence units?

2 Continuous and Matrix View

Let the continuous scene intensity be

$$u \in L^2(\mathbb{R}^2),$$

with object support

$$K = \mathrm{supp}(u).$$

A camera observation is a sampled matrix

$$M = S(u) \in \mathbb{R}^{64\times 64},$$

where

$$S : L^2(\mathbb{R}^2) \to \mathbb{R}^{64\times 64}$$

is a finite sampling operator. Vectorizing the image gives

$$x = \mathrm{vec}(M) \in \mathbb{R}^{4096}.$$

The infinite-dimensional problem is geometric: decide whether the support set K has a property such as convexity, elongation, line-like structure, or can-like cylindrical silhouette. The finite-dimensional detector sees only x, which includes aliasing, missing pixels, quantization, and viewpoint distortion. A practical detector should therefore use regularized evidence rather than exact set-theoretic logic.

Projective transformations are relevant because a phone camera sees objects under perspective. Lines remain lines under projective maps in $PGL(3,\mathbb{R})$, while distances and angles do not. This already favors line-based evidence over raw Euclidean curvature.

3 Approach 1: Radon Line-Contiguity

The Radon representation applies a bank of line integrals. In matrix form,

$$r = A_{\mathrm{Radon}} x,$$

where each row of A_{Radon} sums image evidence along a discretized line tube. In continuous form,

$$\mathcal{R}u(\theta, s) = \int_{\ell_{\theta,s}} u(x, y)\, d\ell.$$

A central fact from convex geometry is:

$$K \text{ is convex} \quad \Longleftrightarrow \quad K \cap \ell \text{ is connected for every line } \ell.$$

Thus convexity can be approximated by checking whether each sampled line hits one foreground interval or multiple separated intervals. Let $c_\ell(t)$ be the number of detected foreground intervals on line ℓ up to time t. A line-contiguity spike can be defined as

$$s_\ell(t) = \mathbf{1}[c_\ell(t) = 1].$$

The accumulated convexity evidence is

$$E_{\mathrm{Radon}}(t) = \sum_{\ell \in \mathcal{L}} s_\ell(t).$$

Classification fires when

$$E_{\mathrm{Radon}}(t) \geq m.$$

3.1 Neuromorphic Fit

The Radon method is an excellent match to spiking hardware:

- each line integral is a synaptic accumulator;
- each orientation and offset can be a separate neuron or neuron group;
- connected-interval detection is a temporal sequence pattern;
- object evidence is a spike count over a line-pattern population;
- projective geometry preserves lines, so the representation is stable under perspective compared with angle- or metric-based features.

For a detector that classifies object n after m patterns over t time steps, Radon evidence has the exact right form:

many weak geometric votes $\to$ thresholded pattern spikes $\to$ class-level spike count.

4 Approach 2: Projective Moment Invariants

Image moments treat intensity as a mass distribution:

$$m_{pq} = \int x^p y^q \, d\mu(x, y).$$

On a matrix,

$$m_{pq} = \sum_x \sum_y x^p y^q M(x, y).$$

This is also linear in the input once the polynomial weights are fixed:

$$\phi = Bx.$$

Scale- and rotation-normalized moments can be formed from central moments. Projective invariants require more algebraic structure, often involving tensor products, determinants, ratios, or cross-ratios:

$$I = \frac{P(m_{pq})}{Q(m_{pq})}.$$

4.1 Neuromorphic Fit

Moment accumulation itself is feasible for an SNN because each moment is a weighted sum. However, projective invariants are less natural because they require:

- multiplication between accumulated quantities;
- division or normalization;
- global integration over the whole object;
- a calibrated classifier in invariant-feature space.

Moments therefore fit best as a secondary stabilizer after a primary pattern-based detector has already localized the candidate object. For example, once Radon/event evidence suggests "small elongated object" or "can-like object," moment features can help reject geometrically inconsistent false positives.

5 Approach 3: Level-Set Hessian Curvature

The level-set approach first embeds a discrete image into a smooth field:

$$u_\sigma = G_\sigma * M,$$

where G_σ is a smoothing kernel. Curvature is then computed from first and second derivatives. For an implicit curve $u_\sigma(x, y) = c$, curvature is

$$\kappa = \frac{u_{xx}u_y^2 - 2u_{xy}u_xu_y + u_{yy}u_x^2}{(u_x^2 + u_y^2)^{3/2}}.$$

Convexity-like evidence can be expressed as sign consistency:

$$S_\kappa = \frac{\max(N_+, N_-)}{N_+ + N_-},$$

where N_+ and N_- are counts of positive and negative curvature samples.

5.1 Neuromorphic Fit

This is the weakest match to a spiking pattern counter. The input event measure μ_t is not differentiable in the classical sense. Derivatives only exist after smoothing:

$$\mu_t \mapsto G_\sigma * \mu_t \mapsto \nabla^2(G_\sigma * \mu_t).$$

The method therefore requires dense reconstruction, convolution, second-order finite differences, and nonlinear division by gradient magnitude. These can be approximated by neural circuits, but they do not naturally produce sparse, independent spike patterns.

6 Comparison

Method	SNN fit	Reason
Radon line-contiguity	Excellent	Linear line accumulation, interval-pattern spikes, direct vote counting, and natural projective robustness because lines map to lines.
Projective moments	Moderate	Moment sums are easy, but projective invariants require global nonlinear algebra and calibration.
Level-set Hessian	Weak	Requires smoothing, second derivatives, curvature ratios, and dense field reconstruction.

Table 1: Fit of each method to an SNN detector that classifies after enough patterns have fired over time.

7 Recommended Architecture

The best architecture is Radon-first with optional moment support:

$$\begin{aligned}\mu_t &\to \text{oriented line/event banks}\\ &\to \text{interval-contiguity neurons}\\ &\to \text{temporal evidence accumulator}\\ &\to \text{class } n.\end{aligned}$$

For object class n, define orientation/offset pattern groups $P_n = \{\ell_1, \dots, \ell_q\}$. Each group detects a line evidence pattern:

$$z_{\ell_i}(t) = \int_0^t \int_\Omega \psi_{\ell_i}(x, y, \tau)\, d\mu.$$

The class fires when

$$\sum_{i=1}^{q} \mathbf{1}[z_{\ell_i}(t) > \theta_i] \geq m_n.$$

Moments can be added as slow global gates:

$$\hat{y} = n \quad \text{if} \quad E_{\text{Radon},n}(t) \geq m_n \quad \text{and} \quad d(\phi(t), C_n) < \epsilon_n,$$

where $\phi(t)$ is a moment feature vector and C_n is the learned feature region for class n.

8 Concrete Example 1: Cigarette Butt Detection at 10–20 m

8.1 Physical and visual assumptions

A cigarette butt in a rural area is a small elongated object, often roughly 20–35 mm long and 5–8 mm wide. At 10–20 m, it is near the lower end of what a phone camera can resolve unless the object has strong contrast against the ground and the camera is stable or moving slowly. The target is not a filled convex blob in practice: it may be partly buried, occluded by grass, broken, or seen as a thin bright/dark segment.

8.2 Neuromorphic pattern design

The detector should not first ask "is this object convex?" It should ask whether enough line-like micro-patterns have accumulated in a small spatial window. A suitable event bank is:

$$\begin{aligned} P_{\text{butt}} = \{ & \text{short parallel line segments}, \\ & \text{two weak long edges}, \\ & \text{high aspect-ratio support}, \\ & \text{filter-tip color contrast}, \\ & \text{temporal persistence under camera motion}\}. \end{aligned}$$

For a candidate window W_c, define oriented Radon evidence:

$$E_\theta(t; W_c) = \sum_s \mathbf{1}[\mathcal{R}_{W_c} \mu_t(\theta, s) > \theta_{\text{line}}].$$

The butt class should fire when a small number of nearly collinear line patterns persist:

$$\sum_{\theta \in \Theta_{\text{local}}} \mathbf{1}[E_\theta(t; W_c) \geq r_\theta] \geq m_{\text{butt}}.$$

8.3 Best method choice

Radon line evidence is the correct primary representation. The object is too small for reliable Hessian curvature, and global projective moments are unstable because a few pixels can dominate the feature vector. Moment features can still help after localization by measuring elongation:

$$\lambda_1 \gg \lambda_2$$

for the second-moment covariance of the candidate pixels.

8.4 Practical detection note

At 20 m, a cigarette butt may occupy only a few pixels in a wide iPhone frame. Reliable detection likely requires one or more of:

- optical zoom or high-resolution capture;
- multiple frames while walking or panning;
- temporal super-resolution from camera motion;
- strong color contrast against soil, asphalt, or grass;
- a low false-positive threshold followed by closer confirmation.

The SNN framing is useful because it can accumulate weak evidence over many frames:

$$E_{\mathrm{butt}}(T) = \sum_{t=1}^{T} \sum_{j \in P_{\mathrm{butt}}} \mathbf{1}[z_j(t) > \theta_j].$$

The classification should occur only after persistent evidence, not from one single frame.

9 Concrete Example 2: Soda Can Detection at 10–20 m

9.1 Physical and visual assumptions

A soda can is much larger than a cigarette butt, typically about 66 mm in diameter and 120 mm tall. At 10–20 m it is still small in a wide phone frame, but it has more usable structure: vertical side edges, elliptical top or bottom arcs, specular highlights, saturated label colors, and a roughly cylindrical silhouette. A crushed can may be non-convex, while an upright or sideways can often has a strong convex hull.

9.2 Neuromorphic pattern design

The class should be represented by a population of line and arc-like evidence patterns:

$$\begin{aligned} P_{\mathrm{can}} = \{&\text{vertical or oblique side-edge pairs,} \\ &\text{top/bottom elliptical arc fragments,} \\ &\text{filled hull contiguity,} \\ &\text{color/reflectance patches,} \\ &\text{temporal stability across frames}\}. \end{aligned}$$

The Radon layer supplies long-edge and hull-contiguity evidence. Let E_{side} count pairs of approximately parallel line responses:

$$E_{\mathrm{side}}(t) = \sum_{\theta, s_1, s_2} \mathbf{1}[\mathcal{R}\mu_t(\theta, s_1) > \theta_e \wedge \mathcal{R}\mu_t(\theta, s_2) > \theta_e \wedge |s_1 - s_2| \in D_{\mathrm{can}}].$$

Let E_{hull} count line-contiguity agreement:

$$E_{\mathrm{hull}}(t) = \sum_{\ell} \mathbf{1}[c_\ell(t) = 1].$$

The class fires when enough can-specific evidence is accumulated:

$$E_{\mathrm{side}}(t) + \alpha E_{\mathrm{hull}}(t) + \beta E_{\mathrm{color}}(t) \geq m_{\mathrm{can}}.$$

9.3 Best method choice

Radon line-contiguity is again the best primary method. It can detect the parallel side boundaries and test whether the filled support behaves like a single connected hull. Moment features are more useful for cans than for cigarette butts because the object is larger and its global aspect ratio and mass distribution are more stable:

$$\phi_{\text{can}} = (\eta_{20}, \eta_{02}, \eta_{11}, h_1, \ldots, h_7).$$

These features can reject non-can clutter after the Radon layer proposes a candidate.

The Hessian method remains a poor primary fit. It may help if the camera is close enough to resolve smooth can boundaries, but at 10–20 m derivative estimates are dominated by aliasing, compression, grass, shadows, and motion blur.

9.4 Practical detection note

Compared with cigarette butts, soda cans are realistic targets at 10–20 m if the object is not heavily occluded. The preferred detector is:

$$\text{Radon side-edge evidence} + \text{hull contiguity} + \text{color/reflectance gating} + \text{temporal persistence}.$$

The SNN should use time to reduce false positives from rocks, leaves, and shiny debris:

$$\hat{y} = \text{can} \quad \text{only if} \quad E_{\text{can}}(T) \geq m_{\text{can}} \text{ for several adjacent windows.}$$

10 Final Conclusion

For a detector that works by firing object class n after m patterns over t time steps, the best match is the Radon transform with line-contiguity patterns. It converts geometric evidence into exactly the kind of sparse, accumulated, thresholded events that spiking networks handle well. Projective moments should be used as a secondary global consistency check, especially for larger objects such as soda cans. Level-set Hessian curvature is mathematically interesting but operationally mismatched to sparse neuromorphic detection, especially for small rural litter at 10–20 m.